

A Sparse-Sensing Spatial Digital Twin for Learning Environmental Impacts of Non-Networked Appliances in a Single Room

Yun-You Lin

Department of Computer Science and Information Engineering
National Changhua University of Education
Changhua, Taiwan
yunitrish0419@gmail.com

Chang-Pei Yi

Department of Computer Science and Information Engineering
National Changhua University of Education
Changhua, Taiwan

Hui-Yu Shen

Department of Electrical Engineering
Chienkuo Technology University
Changhua, Taiwan

Abstract—Practical room-scale digital twins must often operate with sparse sensors and non-networked appliances that expose no telemetry. This paper presents a single-room spatial digital twin that estimates temperature, humidity, and illuminance fields from eight corner sensors while modeling conventional air conditioners, manual windows, and ordinary lights. The proposed design combines an appliance-aware bulk-plus-local field model, photometric lamp falloff, lightweight one-bounce diffuse illuminance reflection, active-device power calibration, trilinear residual correction, least-squares appliance-impact learning, and an optional hybrid residual neural layer. Across eight canonical scenarios, the base estimator achieves average field MAE of 0.0474/0.1765/2.0269 for temperature, humidity, and illuminance, compared with 0.1723/0.4633/54.9052 for IDW. A leave-one-scenario-out hybrid residual evaluation further reduces MAE to 0.0017/0.0059/0.1407. On a preliminary seven-day bedroom snapshot dataset, calibration reduces held-out pillow-point MAE from 0.8967/4.1286/309.0142 to 0.1676/0.3939/16.6450, with positive date-block bootstrap intervals and date-deletion reductions for all three factors. The evaluation separates controlled full-field evidence, public task-aligned comparisons, and real sparse-calibration checks. These results indicate that sparse corner sensing can support an interpretable and trainable indoor twin when physical structure, calibration, and residual learning are assigned complementary roles.

Index Terms—spatial digital twin, sparse sensing, indoor environment modeling, non-networked appliances, reduced-order model, sensor calibration

I. Introduction

Smart homes and smart buildings require indoor environmental awareness for comfort assessment, energy management, and device control. A practical system should estimate conditions at relevant regions such as a room center, window-side zone, or bed location, not only at installed sensors. This is difficult because dense sensing is intrusive, while ordinary rooms also contain conventional

appliances that do not report state, power, or operating parameters.

This work targets a common but under-instrumented digital-twin setting: the room contains influential physical assets, but those assets are not necessarily connected to a building management system. The twin must therefore infer appliance state effects from sparse environmental observations, support what-if queries, and keep its claim boundary clear. Pure interpolation ignores appliance geometry and directionality, while a purely local field model misses whole-room settling after air-conditioner activation. We therefore use a reduced-order appliance-aware model as the primary estimator, sensor residuals for calibration, and neural learning only for structured residual correction rather than for an end-to-end black-box field predictor.

The contributions are: (i) a three-factor room digital twin with bulk-plus-local appliance influence functions, photometric lamp falloff, and one-bounce diffuse illuminance reflection; (ii) a sparse-sensor calibration pipeline combining active-device power fitting, trilinear residual correction, and before/after impact learning; (iii) a hybrid residual pathway that preserves the reduced-order estimator while refining systematic error; and (iv) an implementation and validation hierarchy covering synthetic full-field scenarios, ablations, leave-one-scenario-out residual learning, window-condition cases, public task-aligned datasets, and preliminary real-bedroom snapshots. Figure 1 summarizes the argument chain from the research gap through methods and evidence to the claim boundary; it also keeps the MCP-compatible interface secondary to the modeling contribution and marks intervention-based recommendation validation as future work.

Overall Research Logic Architecture

From research gap and RQ1-RQ4 through methods and E1-E9 evidence to bounded conclusions

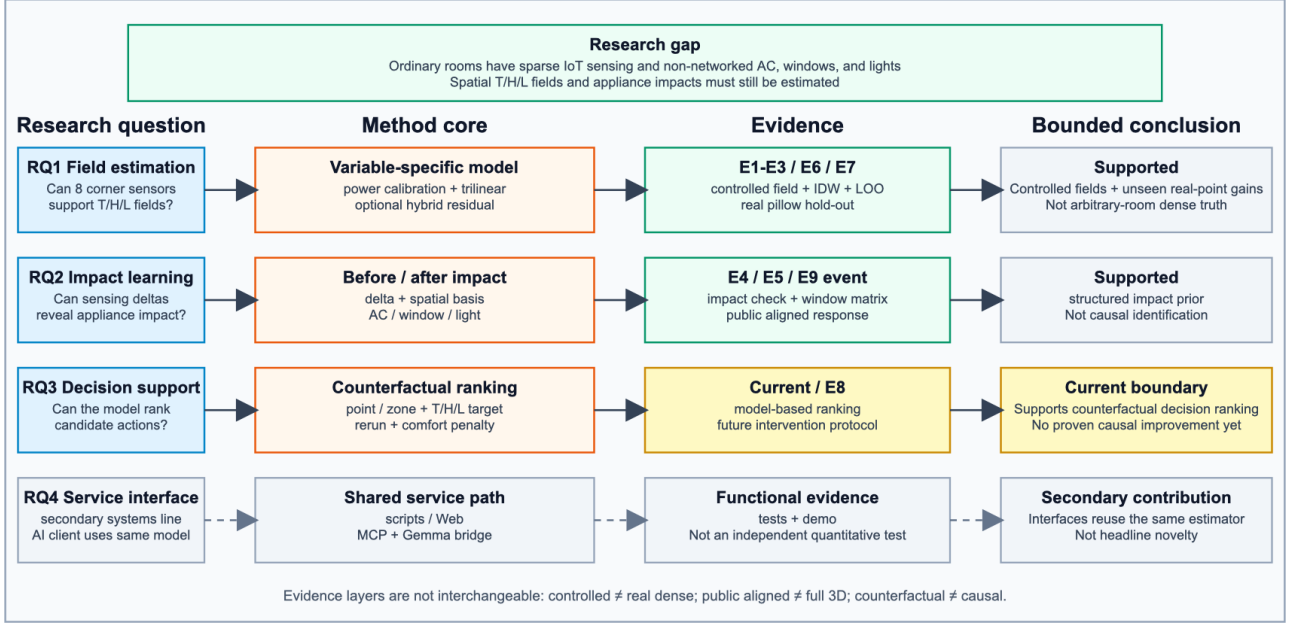


Fig. 1. Overall argument map linking the sparse-sensing and non-networked-appliance gap to RQ1–RQ4, method components, the E1–E9 evidence hierarchy, and bounded conclusions. RQ4 is a secondary service line, while E8 remains a future intervention protocol.

II. Related Work

Digital twins provide computational representations of physical systems and are increasingly used in manufacturing, infrastructure, and smart buildings [1], [2]. Building-oriented reviews show strong attention to BIM/BMS connectivity and system architecture, but sparse-sensor spatial estimation in ordinary rooms remains less explored [3]. For control-oriented indoor modeling, reduced-order and grey-box approaches remain attractive because they are interpretable and tractable compared with CFD [4]–[6]. Zonal and hybrid indoor models preserve dominant spatial non-uniformity at lower cost [7]–[9]. Simulation-informed hybrid forecasting has also used forecast-day physical outputs as a baseline and learned the historical simulation–measurement gap for next-day indoor-temperature prediction [10]; a vanilla recurrent comparator follows the Elman formulation [11]. Data-assimilation studies show that finite observations can reconstruct indoor temperature and humidity fields and that sensor placement matters [12], [13]. Controlled-growth studies further motivate dynamic rather than strictly constant temperature, humidity, and light programs [14], while greenhouse Kalman studies report both successful online adaptation and filtering that deteriorated monitoring when model accuracy was insufficient [15], [16]. The present work combines these strands by coupling sparse sensing with appliance-aware influence bases and residual calibration. IDW [17] is used as a geometry-free interpolation baseline. The service

interface is secondary to the modeling contribution, but the same estimator is exposed through local APIs, a web demo, and an MCP-compatible tool layer [18].

III. Method

A. Room, Devices, and Field Model

The canonical room is a 6.0 m by 4.0 m by 3.0 m rectangular room with eight corner sensing nodes, four on the floor and four on the ceiling. Each node measures temperature, humidity, and illuminance. The room contains three evaluation zones and three appliance classes: an air conditioner, a window, and a light. The web demo represents these appliances as wall or ceiling markers and allows their activations to be toggled.

For metric $m \in \{T, H, L\}$ at point $\mathbf{p} = (x, y, z)$ and elapsed time t , the estimated field is

$$F_m(\mathbf{p}, t) = B_m^{\text{bulk}}(t) + B_m^{\text{local}}(\mathbf{p}, t) + \sum_j I_{j,m}(\mathbf{p}, t) + C_m(\mathbf{p}). \quad (1)$$

The bulk term captures whole-room settling, local terms preserve spatial gradients, $I_{j,m}$ represents appliance influence, and C_m is fitted from sparse sensor residuals. Each appliance influence is modulated by

$$E_j(\mathbf{p}, t) = a_j(1 - e^{-t/\tau_j})D_j(\mathbf{p})G_j(\mathbf{p}), \quad (2)$$

where a_j is activation, τ_j is a response time constant, D_j is distance attenuation, and G_j is directional gain. The air conditioner lowers temperature and humidity, the window

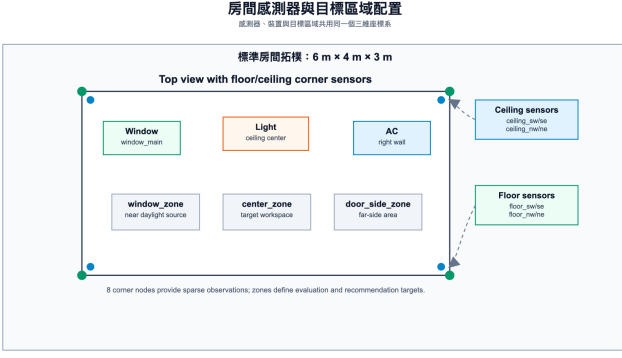


Fig. 2. Room topology, appliance placement, target zones, and the eight floor/ceiling corner sensors.

uses outdoor temperature, humidity, and sunlight, and the light primarily raises illuminance with a small heat gain. The lamp direct term uses a cosine beam weight derived from beam angle, an inverse-square distance scale with bounded gain, and line-of-sight obstruction. Window daylight remains an envelope-based daylight term in the reported experiments; an aperture view-factor option is implemented but not used as the default because it increased window-family illuminance error in the current canonical setup.

Direct window and lamp terms underestimate indirect fill light near floors, walls, and furniture. Instead of full ray tracing, illuminance adds a lightweight one-bounce diffuse term

$$I^{\text{refl}}(\mathbf{p}, t) = \sum_{s \in \mathcal{S}} \rho_s \bar{I}_s A_s^{\text{rel}} e^{-\|\mathbf{p} - \mathbf{c}_s\|/\ell_s} \gamma_s(\mathbf{p}) V_s(\mathbf{p}), \quad (3)$$

where surfaces $\mathcal{S}$ include room boundaries and active furniture faces, ρ_s is reflectance, $\bar{I}_s$ is direct illuminance at surface center $\mathbf{c}_s$, A_s^{rel} is normalized area, ℓ_s is attenuation length, γ_s is diffuse alignment, and V_s is an obstruction term.

B. Sparse Calibration and Learning

Given predicted sensor value $\hat{y}_{i,m}$ and observed value $y_{i,m}$, residual $r_{i,m} = y_{i,m} - \hat{y}_{i,m}$ is used in two steps. First, active-device power scales are fitted by least squares over sensor residuals. Second, a trilinear correction is fitted:

$$C_m(x, y, z) = c_{0,m} + c_{1,m}X + c_{2,m}Y + c_{3,m}Z + c_{4,m}XY + c_{5,m}XZ + c_{6,m}YZ + c_{7,m}XYZ, \quad (4)$$

where X, Y, Z are normalized room coordinates. This cannot recover high-frequency turbulence, but it corrects bias, first-order gradients, and corner interactions from the eight observations. The trilinear form is deliberately matched to the sensing layout. With floor and ceiling sensors at all four horizontal corners, the eight observations constrain the eight coefficients without introducing a high-order surface that would overfit sparse points. The correction should therefore be interpreted as a low-frequency

bias and gradient adjustment, not as a substitute for dense flow or lighting measurements. This design keeps the estimator identifiable from the available observations while still letting appliance geometry and dynamics remain in the explicit physical layer.

For non-networked devices, before/after sensor deltas supervise impact learning. The learning event is stored as an appliance-state record rather than as a free-text action label: it includes a record id, device name, applied device state, merged runtime device specifications, indoor baseline, outdoor boundary conditions, furniture state, elapsed time, and before observations. For an air conditioner, the applied state may include activation, mode, setpoint, fan speed or strength, outlet angles, and sweep settings. After observations from the same sensors complete the record. Given metric deltas $\Delta \mathbf{y}_m$ and appliance envelopes in matrix X , coefficients are estimated by

$$\beta_m = \arg \min_{\beta} \|\Delta \mathbf{y}_m - X\beta\|_2^2. \quad (5)$$

For residual learning, point features include normalized coordinates, elapsed time, indoor/outdoor conditions, base estimates, device activations, calibrated powers, appliance envelopes, and air-conditioner operating state such as mode, setpoint, fan strength, outlet angles, and sweep settings. The hybrid residual layer predicts $R_m(\mathbf{p}, t; \theta_m)$ and returns $F_m^{\text{hybrid}} = F_m + R_m$. This is the current-state spatial form used in the reported experiments. Its time-aligned h -step generalization is

$$\hat{F}_m^{\text{hybrid}}(\mathbf{p}, t+h | \mathcal{I}_t) = \hat{F}_m^{\text{phys}}(\mathbf{p}, t+h | \mathcal{I}_t) + \hat{R}_m(\mathbf{p}, t+h | \mathcal{I}_t). \quad (6)$$

Here, h is forecast lead rather than a physical coefficient. The physics term carries $t+h$ because it is propagated from information available at forecast origin t to produce the baseline at the same target time as the residual forecast. The information set $\mathcal{I}_t$ may include forecast boundary conditions and planned controls but excludes observations or truth residuals from $t+h$. The present spatial-estimation experiments correspond to $h=0$ and do not constitute a new multi-step forecasting evaluation. Temperature and humidity residual traces can be Fourier low-pass filtered before training because thermal and moisture responses are typically smoothed by heat capacity, air mixing, moisture exchange, and dehumidification. Illuminance uses raw residual labels because lighting, daylight, shadows, occlusion, and reflection can create physically meaningful short-timescale changes that low-pass filtering would remove.

The data flow in Fig. 3 is intentionally shared between training and runtime use. During learning, raw corner-sensor records, device events, outdoor boundary conditions, and scenario metadata are first time-aligned and normalized, then assembled into the same scenario state used by the service layer. This state is evaluated by the reduced-order estimator to produce base fields and

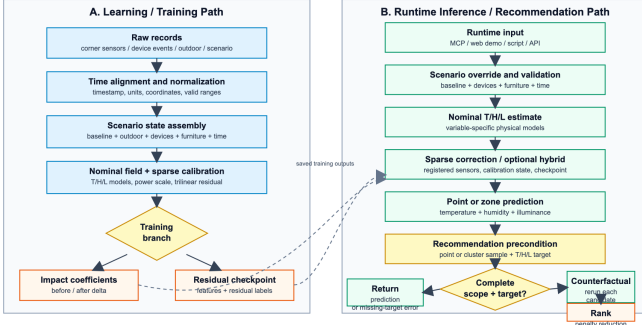


Fig. 3. Shared learning, inference, and recommendation data flow. Learned impact coefficients and the optional residual checkpoint are saved from the training path and reused by the runtime estimator.

residual labels. The pipeline then branches into least-squares impact learning from before/after sensor deltas and optional hybrid residual training from point-level residual samples, producing learned coefficients, validation summaries, and a checkpoint. At runtime, a user, web interface, or MCP client supplies the current baseline, outdoor conditions, device and furniture states, elapsed or steady-state time, and a query point or target zone. The service rebuilds the scenario state, evaluates the nominal fields, applies sparse correction and the optional residual checkpoint, and returns point or zone estimates. Action ranking is defined only after a decision sample scope is specified, either a single query point or a cluster/zone sample, and after a complete three-factor temperature, humidity, and illuminance target/tolerance vector is supplied. Without those inputs, the service returns estimates only and does not emit recommendations. Given scope and target, each candidate device action is converted into a counterfactual scenario and passed through the same inference path; actions are sorted by predicted comfort-penalty reduction.

Candidate control actions are evaluated as model-based counterfactuals. Let $S = \{\mathbf{p}_k\}_{k=1}^K$ be the decision sample scope, where $K = 1$ is a point query and $K > 1$ is a cluster or target-zone sample. For metric m , the scope value is $q_m(S) = K^{-1} \sum_k F_m(\mathbf{p}_k, t)$. A weighted comfort penalty is computed from normalized deviations beyond target tolerances for temperature, humidity, and illuminance, and each action is ranked by current penalty minus simulated post-action penalty. AC actions carry a fuller operating state rather than only an on/off flag: cooling, drying, heating, or fan mode; temperature setpoint; fan speed or numeric fan strength; horizontal and vertical outlet angles; and fixed or sweeping airflow. This is recommendation ranking, not closed-loop control; causal validation requires before/after intervention trials.

IV. Implementation and Experimental Setup

The prototype is implemented as a zero-dependency Python package with modules for entities, scenarios, field simulation, impact learning, hybrid residual training, IDW comparison, action ranking, service APIs, an MCP endpoint, and a rotatable web demo. The shared service layer still supports the reproduction scripts and browser demo, while the MCP endpoint is intentionally narrowed to five runtime tools: environment initialization, elapsed-time or steady-state point sampling, before/after impact-learning records, direct window input, and point-target action ranking from registered devices. The ranking endpoint requires the point coordinates plus a complete temperature, humidity, and illuminance target; missing target fields produce no recommendation. The initialization tool registers the runtime state used by subsequent calls, including the base scenario, indoor baseline, outdoor boundary conditions, device and furniture states, AC mode/setpoint/fan/airflow controls when present, default elapsed or steady-state timing, and the estimator choice.

The service layer is intentionally kept separate from the estimator. A command-line script, the MCP endpoint, and the browser demo call the same scenario and sampling functions, so evaluation results are not produced by a separate code path from the interactive prototype. The browser interface supports metric switching, appliance toggling, AC mode/setpoint/fan-speed/outlet-angle controls, indoor baseline editing, season-weather-time presets, outdoor boundary overrides, point sampling, target-zone summaries, impact-learning output, and estimator switching between the base and hybrid residual fields when a saved checkpoint is available. This separation also reduces an implementation risk common in demonstration-driven digital twins: the visual interface can change without changing the numerical estimator, and the estimator can be tested from scripts without relying on browser state. The MCP layer is consequently treated as a deployment interface for the same service functions rather than as a separate modeling contribution. In the present paper, this matters because reported MAE values remain tied to reproduction scripts, while MCP point samples, learning records, direct window inputs, and action rankings expose the same runtime estimator without turning validation utilities such as IDW comparison or the full window matrix into MCP-facing claims.

All synthetic full-field experiments use the canonical room and $16 \times 12 \times 6$ grid, giving 1152 spatial samples per field. Scenarios are evaluated after an 18-minute settling interval with outdoor temperature 33.0°C, relative humidity 74.0%, and sunlight 32000 lux. The eight canonical scenarios are idle, AC only, window only, light only, AC+window, window+light, AC+light, and all active. Dense synthetic truth is generated by deterministic appliance power adjustments and sensor observations are synthesized from truth sensor

predictions with fixed index-based perturbations: temperature adds $0.08((i \bmod 4) - 1.5)$, humidity adds $0.3((i \bmod 4) - 1.5)$, and illuminance adds $3.0((i \bmod 4) - 1.5)$. The main reproduction scripts are `run_demo.py`, `run_hybrid_residual_experiment.py`, `run_submission_readiness_experiments.py`, `run_bedroom_weekly_simulation.py`, and `run_public_dataset_model_comparison.py`.

The bounded indoor operating and application domain considered in this study is 20–30°C. Outdoor boundary inputs may lie outside that interval, as in the canonical hot-weather case, but they do not extend the claimed range of indoor target states or candidate deployment. Human-comfort ranking therefore uses target tolerances rather than assuming that smaller numerical error proves a need for equally narrow actuation.

Because no single dataset covers room geometry, dense 3-D ground truth, appliance state, and real sparse observations at once, the evaluation uses the layered claim boundary in Table I. We label the validation items as E1–E9 so that each numerical claim is tied to one evidence layer: E1–E6 are controlled synthetic or robustness experiments, E7 is a real-bedroom sparse-calibration check, E8 is a protocol for future before/after recommendation validation, and E9 is the public task-aligned benchmark. This separation is important because good performance on one layer does not automatically validate the others.

V. Evaluation

A. E1–E4 Synthetic Reconstruction and Impact Checks

Across the eight canonical scenarios, the base estimator achieves average field MAE of 0.0474 for temperature, 0.1765 for humidity, and 2.0269 for illuminance. IDW gives 0.1723, 0.4633, and 54.9052, respectively. Table II summarizes the main comparisons and ablations. Removing appliance-aware feedback produces larger temperature and illuminance errors; removing one-bounce reflection or active-device calibration mainly hurts illuminance. The no-trilinear variant can be lower on dense synthetic humidity and illuminance because trilinear correction enforces sparse sensor consistency rather than guaranteed dense-field monotonic improvement. This last point is important for interpreting the ablation table. The trilinear correction is fitted from corner observations, so it is optimized for consistency with sparse sensors and for the real-bedroom calibration setting. It is not optimized directly against every dense grid point in the synthetic room. A lower dense-field MAE in the no-trilinear row therefore does not imply that sparse calibration is unnecessary; it indicates that sparse correction and dense synthetic truth are different objectives. For deployment, the estimator must respect measured corner observations first, then use the appliance-aware model to extrapolate the field between them.

For impact learning, the AC-only scenario learns negative coefficients for temperature and humidity and near-

TABLE I
Validation Scope and Claim Boundary

Evidence item	Supports	Boundary
E1–E4 synthetic full-field, IDW, ablation, and impact-learning checks	3-D reconstruction, baseline comparison, and component contribution under controlled truth	Simulated room and deterministic perturbations
E5 48-case window matrix	Sensitivity to outdoor temperature, humidity, sunlight, and opening ratio	34 in-domain cases; 14 out-of-domain stress cases
E6 hybrid residual splits	Residual learnability under default, no-Fourier, and leave-one-scenario-out settings	Scenario-family robustness, not arbitrary-room generalization
E7 seven-day bedroom snapshots	Sparse calibration improves a held-out pillow point	One reference point; no dense real 3-D ground truth
E8 recommendation intervention protocol	Defines measured improvement, success rate, and top-1 regret	Protocol only; not counted as completed validation
E9 SML2010 and CU-BEMS task-aligned benchmarks	External positioning on compatible time-series or zone-level prediction tasks	Not direct validation of the full spatial twin

Field MAE comparison

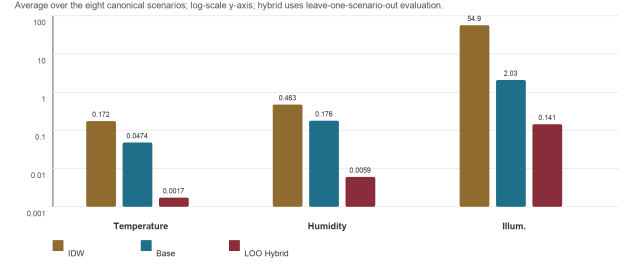


Fig. 4. Average field MAE for IDW, the base estimator, and leave-one-scenario-out hybrid residual correction. The y-axis is logarithmic.

zero illuminance effect. The light-only scenario learns a strongly positive illuminance coefficient, weak positive heat gain, and near-zero humidity effect. These directions match the intended physical interpretation of the appliances.

B. E6 Hybrid Residual Correction

The default held-out residual experiment trains on six scenarios and tests on `light_only` and `all_active`, yielding 576 training and 192 test samples. With Fourier denoising for temperature and humidity residual traces, hybrid MAE is 0.0020/0.0051/0.1370, compared with base MAE of 0.0474/0.1765/2.1757 on the same split. Illuminance is intentionally left unfiltered because its fast changes

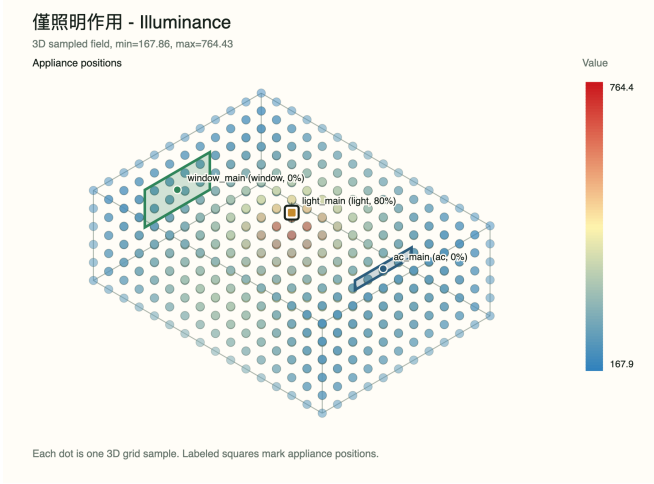


Fig. 5. Example 3-D illuminance field for the light-only scenario, showing localized lighting influence and reflected fill.

TABLE II
Synthetic Field MAE and Robustness Summary

Setting	Temp.	Hum.	Illum.
IDW baseline	0.1723	0.4633	54.9052
Raw nominal	0.1312	0.0842	3.5183
No reflection	0.0472	0.1762	2.4296
No calibration	0.0493	0.1772	3.3631
No trilinear	0.0446	0.0274	0.9849
Full base	0.0474	0.1765	2.0269
LOO hybrid avg.	0.0017	0.0059	0.1407

can correspond to source switching, daylight boundary changes, shadows, and reflections rather than noise. Disabling Fourier gives 0.0021/0.0057/0.1370, showing that the illuminance improvement is not caused by frequency-domain smoothing. In leave-one-scenario-out evaluation, each fold trains on seven scenarios and tests on one scenario, with 672/96 train/test samples. The average LOO hybrid MAE is 0.0017/0.0059/0.1407, corresponding to approximately 96.41%, 96.66%, and 93.06% reductions relative to the base estimator.

C. E7 Real-Bedroom Snapshot Validation and E8 Intervention Protocol

We further evaluate calibration on a user-supplied bedroom dataset. The room is a 4.0 m by 4.6 m by 3.2 m bedroom with wall-mounted AC, southeast-facing window, ceiling light, desk lamp, bed, desk, and cabinets. The dataset covers 2026-04-14 to 2026-04-20, with 09:00, 15:00, 22:00, and 02:00 snapshots each day, for 28 cases. Each case provides eight corner observations, appliance activations, outdoor boundary information, and a pillow-location reference observation. Corner sensors are used

TABLE III
Real-Bedroom Snapshot Comparison

Case	Temp.	Hum.	Illum.	Penalty
Raw pillow MAE	0.8967	4.1286	309.0142	—
Calib. pillow MAE	0.1676	0.3939	16.6450	0.0911
Morning calib.	0.2372	0.3695	8.6247	0.2371
Afternoon calib.	0.1358	0.4890	11.2235	0.0812
Night calib.	0.0913	0.2193	46.6993	0.0459
Sleep calib.	0.2060	0.4979	0.0326	0.0000

for active-device calibration and trilinear correction; the pillow point is held out.

Table III shows that calibration reduces pillow-point MAE from 0.8967°C, 4.1286% relative humidity, and 309.0142 lux to 0.1676°C, 0.3939%, and 16.6450 lux. A 20,000-replicate paired date-block bootstrap (seed 20260726) yields mean reductions and 95% intervals of 0.7291 [0.4582, 1.0232]°C, 3.7346 [3.2005, 4.2524]%RH, and 292.3692 [288.3083, 297.0237] lux; 26/28, 28/28, and 28/28 snapshots improve. Seven overlapping leave-one-date-out folds retain positive minimum reductions of 0.6123°C, 3.5551%RH, and 290.5716 lux. This is an internal sensitivity check over one room, seven dates, and one held-out point, not independent replication or intervention success. Sleep snapshots use a 0 lux illuminance target with 5 lux tolerance so dark sleep conditions are not penalized. These snapshots validate sparse-sensor calibration and point-level estimation, not recommendation efficacy. A versioned E8 trial schema, empty field template, and deterministic analyzer now preregister the before/after endpoints. E8 currently has 0 completed real intervention trials; its summary remains NOT_EVALUATED, and all efficacy estimates remain null.

D. E5 Window Matrix and E9 Public Benchmarks

The window matrix contains 48 cases from four times of day, three weather states, and four seasons. It evaluates the window as an environmental variable rather than a binary switch, and the direct-input mode accepts outdoor temperature, outdoor humidity, sunlight, and opening ratio without discretizing the outside condition. Under the current indoor 20–30°C domain, 34 target-zone results are in range and 14 are retained only as out-of-domain stress cases. Representative window-zone illuminance values are 73.08 lux for winter-rain-night, 96.80 for spring-cloud-morning, and 247.30 for summer-sun-noon.

For external comparison, SML2010 [19] and CU-BEMS [20] support only task-aligned time-series or zone-level tests because neither supplies matched room geometry and dense 3-D truth. The proposed model is strongest on event-delta and longer-horizon temperature tasks, but persistence remains strong for short-window illuminance and high-inertia CU-BEMS zones; no CU-BEMS task beats persistence. Table IV retains representative favorable and adverse cases. An Oh et al.-inspired fixed ridge residual transfer [10] is useful at short horizons but does not reproduce the confidential CNN-LSTM or its

TABLE IV
Selected Task-Aligned Benchmark Results

Task	Metric	Pers.	LinReg	Proposed
S3 15min temp.	MAE	0.233	0.093	0.071
S3 15min temp.	Corr.	0.000	0.909	0.950
S3 60min temp.	MAE	0.563	0.212	0.190
S3 60min illum.	RMSE	21.93	22.76	19.34
S1 15min illum.	MAE	3.42	4.02	5.35
C2 15min illum.	MAE	1.36	1.79	7.70
S2 15min hum.	MAE	0.20	0.21	0.83

next-day advantage. A preregistered next-day follow-up also worsened MAE to 1.629/1.625 versus persistence at 1.517/1.500, with intervals crossing zero. The RNN same-data comparison is complete: the fixed model used the same four-record histories, splits, targets, and test rows in all 12 cases; linear regression was lowest in 7, persistence in 5, and RNN in 0. Thus the recurrent comparison provides no superiority claim.

E. Validity Considerations

Dense full-field truth is synthetic; robustness checks do not create measured dense truth. E7 covers one room, seven dates, and one held-out point, while public data validate only aligned subtasks. The fixed RNN does not rule out other recurrent designs. Application claims are limited to indoor states within 20–30°C; dynamic plant growth remains a candidate because the model lacks PPFD/PAR, CO₂, substrate, airflow, and biological endpoints. Action rankings remain counterfactual pending interventions.

VI. Discussion and Conclusion

The appliance-aware bulk-plus-local model outperforms geometry-free interpolation under controlled truth, while residual calibration improves the sparse real-room point under date-block resampling. Public data support aligned subtasks, not arbitrary-building generalization or causal control. Future work will add longer deployment, dense measured truth, interventions, humidity alignment, and a preregistered same-row unfiltered/moving-average/KF/EKF comparison after defining the state, observation, and covariance models; Kalman performance is not yet evaluated.

Acknowledgment

The authors thank the Department of Computer Science and Information Engineering, National Changhua University of Education, for supporting the thesis prototype development.

References

- [1] F. Tao, H. Zhang, A. Liu, and A. Y. C. Nee, “Digital twin in industry: State-of-the-art,” *IEEE Transactions on Industrial Informatics*, vol. 15, no. 4, pp. 2405–2415, 2019.
- [2] A. Fuller, Z. Fan, C. Day, and C. Barlow, “Digital twin: Enabling technologies, challenges and open research,” *IEEE Access*, vol. 8, pp. 108 952–108 971, 2020.

- [3] A. S. Cespedes-Cubides and M. Jradi, “A review of building digital twins to improve energy efficiency in the building operational stage,” *Energy Informatics*, vol. 7, p. 11, 2024.
- [4] P. Bacher and H. Madsen, “Identifying suitable models for the heat dynamics of buildings,” *Energy and Buildings*, vol. 43, no. 7, pp. 1511–1522, 2011.
- [5] P. Hietaharju, M. Ruusunen, and K. Leiviskä, “A dynamic model for indoor temperature prediction in buildings,” *Energies*, vol. 11, no. 6, p. 1477, 2018.
- [6] G. Gokhale, B. Claessens, and C. Develder, “Physics informed neural networks for control oriented thermal modeling of buildings,” *Applied Energy*, vol. 314, p. 118852, 2022.
- [7] E. J. Teshome and F. Haghighat, “Zonal models for indoor air flow - a critical review,” *International Journal of Ventilation*, vol. 3, no. 2, pp. 119–129, 2004.
- [8] A. Megri, Y. Yu, R. Miao, and X. Hu, “A new dynamic zonal model with air-diffuser (doma) - application to thermal comfort prediction,” *Indoor and Built Environment*, vol. 31, no. 7, pp. 1738–1757, 2022.
- [9] B. Huljak, J. A. Acero, Z. H. Kyaw, and F. Chinesta, “Hybrid models for simulating indoor temperature distribution in air-conditioned spaces,” *Frontiers in Built Environment*, vol. 11, p. 1690062, 2025.
- [10] J.-H. Oh, S. Sfarra, and E.-J. Kim, “Hybrid modeling based on integrating simulation and operational data to improve indoor air temperature predictions, a controlled variable in digital twin models,” *Energy and Buildings*, vol. 324, p. 114898, 2024.
- [11] J. L. Elman, “Finding structure in time,” *Cognitive Science*, vol. 14, no. 2, pp. 179–211, 1990.
- [12] W. Qian, C. Li, H. Gao, L. Zhuang, Y. Lu, S. Hu, and J. Liu, “Estimating indoor air temperature and humidity distributions by data assimilation with finite observations: Validation using an actual residential room,” *Building and Environment*, vol. 269, p. 112495, 2025.
- [13] Y. L. Chen and J. Wen, “Application of zonal model on indoor air sensor network design,” in *Sensors and Smart Structures Technologies for Civil, Mechanical, and Aerospace Systems 2007*, ser. Proceedings of SPIE, vol. 6529, 2007, p. 652911.
- [14] C. Chiang, D. B  nkestad, and G. Hoch, “Reaching natural growth: The significance of light and temperature fluctuations in plant performance in indoor growth facilities,” *Plants*, vol. 9, no. 10, p. 1312, 2020.
- [15] S. L. Speetjens, J. D. Stigter, and G. van Straten, “Towards an adaptive model for greenhouse control,” *Computers and Electronics in Agriculture*, vol. 67, no. 1–2, pp. 1–8, 2009.
- [16] S. van Mourik, P. J. M. van Beveren, I. L. L  pez-Cruz, and E. J. van Henten, “Improving climate monitoring in greenhouse cultivation via model based filtering,” *Biosystems Engineering*, vol. 181, pp. 40–51, 2019.
- [17] D. Shepard, “A two-dimensional interpolation function for irregularly-spaced data,” in *Proceedings of the 1968 ACM National Conference*, 1968, pp. 517–524.
- [18] Model Context Protocol, “Model context protocol documentation,” <https://modelcontextprotocol.io/>, accessed: 2026-04-10.
- [19] P. Romeu-Guallart and F. Zamora-Martinez, “SML2010 [dataset],” *UCI Machine Learning Repository*, 2014.
- [20] G. Chitalia and M. Pipattanasomporn, “CU-BEMS, smart building electricity consumption and indoor environmental sensor datasets,” *Scientific Data*, vol. 7, p. 290, 2020.